Adult Pharmacological Agents: Master List
G-6
Effective: May 30, 2020

THIS MASTER LIST REFLECTS ONLY THE PROTOCOL AND DOSING FOR THE FOLLOWING FORMULARY OF MEDICATIONS CARRIED BY FFEMS. IT DOES NOT REFLECT ALL THE INDICATIONS, CONTRAINDICATIONS, PRECAUTIONS AND USAGE DIRECTIONS FOR THE MEDICATION. YOU MUST REFER TO THE PROTOCOL FOR THIS INFORMATION.

Adenosine: *Protocol: C-10 Supraventricular Tachycardia*

IF RHYTHM IS NOT ATRIAL FIBRILLATION administer adenosine 12 mg rapid IV push over 1-3 seconds to be followed by a 20 ml saline flush. If no response, may repeat in 1-2 minutes at 12 mg IV push over 1-3 seconds followed by 20 ml saline flush.

Albuterol: *Protocol: M-7 Respiratory Distress With/Without Wheezes*

Albuterol 2.5 mg in a prediluted solution as an updraft through the nebulizer

Amiodarone: *Protocol: C-9 Ventricular Fibrillation/Pulseless Ventricular Tachy*cardia

Following initial epinephrine, administer Amiodarone 300 mg IV or IO if patient continues in VF or pulseless VT at next rhythm check. This may be repeated in 3-5 minutes at a dose of 150 mg IV or IO.

Protocol: C-11 Undifferentiated Wide Complex Tachycardia

If Amiodarone is ordered "AT MD DISCRETION ONLY", administer **Amiodarone 150 mg** in 100 ml D5W over ten minutes.

Atrovent: *Protocol: M-7 Respiratory Distress With/Without Wheezes*

If second treatment is needed based on assessment, administer neb in combination with Atrovent 500 mcg (2.5 ml prediluted solution) as an updraft through the nebulizer.

Aspirin: *Protocol: C-4 chest Pain believed to be Cardiac in Origin*

If the patient has no contraindication, administer four (4) chewable baby aspirin, 81 mg each, total dose 324 mg

Atropine: *Protocol: C-3 Bradycardia*

Administer atropine 0.5 mg IV push rapidly initially. Repeat atropine 0.5mg IV push rapidly in 3- 5 minutes if needed. If rhythm continues, administer atropine 1 mg IVP rapidly.

Protocol: O-2 Suspected Overdose/Poisoning

Administer atropine 1-2 mg IV push. May be repeated every five minutes.

Benadryl: *Protocol: M-2 Allergic Reaction/Anaphylaxis*

Benadryl 25 mg IM or IVP slowly.

Protocol: O-2 Suspected Poisoning/Overdose

Administer Benadryl 50 mg IV or IM

Calcium Chloride: *Protocol: O-2 Suspected Overdose/Poisoning*

Administer Calcium Chloride 10% 1 Gram at a rate of 1 ml/minute for calcium channel or beta blocker overdose

Protocol: C-7 PEA/Asystole

Administer 1 G (1 amp) 10% Calcium Chloride IVP/IO if hyperkalemic. or suspected calcium channel or beta blocker overdose.

Cyanokit: *Protocol: O-2 Suspected Overdose/Poisoning*

Administer Cyanokit (Hydroxocobalamine) 5 g IVPB over 15 minutes into its own dedicated IV line if suspected cyanide exposure.

Dextrose: Protocol: M-3 Hypo/Hyperglycemia

Administer **12.5 to 25g of dextrose** IV/IO. Dose may be repeated once in 5 minutes if no or limited response.

Protocol: C-7 PEA/Asystole

If glucose is below 60 mg/dL, administer 25 grams of **dextrose** to the adult IV/IO.

Epinephrine 1:1,000: *Protocol: M-2 Allergic Reaction/Anaphylaxis*

BLS: Epinephrine 1:1000 concentration IM into the lateral or anterior thigh:

- Children > 60 lbs. and Adults < 65 years old administer Epinephrine 0.3 mg IM
  Further dosing MUST notify Medical Control.
- Children 25 lbs. to 60 lbs. and Adults ≥ 65 years of age administer Epinephrine 0.15 mg IM
  Repeat Epinephrine at same dose if no improvement in symptoms for the Adult ≥ 65 years of age.
- Children < 25 lbs. CALL MEDICAL CONTROL

ALS: For adults ≥ 65 years old with history of cardiac disease, take Beta-Blockers / Digoxin or patients who have heart rates ≥ 150 in moderate distress, Epinephrine (0.15 mg of 1:1000) IM. Repeat in 5 minutes if no improvement in symptoms.

- All other adults in moderate distress consider Epinephrine 0.3 mg to 0.5 mg IM. Repeat in 5 minutes ONLY if respiratory distress continues.
- Adults in severe distress, Administer medications as in Moderate symptoms. Repeat epinephrine at same dose in 5 minutes if no improvement in symptoms. Additional doses AT MD DISCRETION ONLY.

Protocol: T-1 Burns

If rapidly swelling shut and intubation is not likely, nebulize 1:1,000 epinephrine 3 ml mixed with 3ml of saline.

Protocol: M-7 Respiratory Distress with/without Wheezing

For stridor, nebulize 3ml of epinephrine 1:1,000 with 3 ml of saline

Protocol: T-5 Hypotension and Shock Adult and Pediatric

- If B/P < 90 systolic **WITH signs of poor perfusion** (ie. Mottled skin, decreasing LOC, capnography < 25) administer a 500 ml IV fluid bolus while preparing push dose epinephrine.
- **Push Dose Epinephrine is prepared** by placing 1mg of 1:10,000 or 1:1,000 epinephrine in 100 ml NS. This reconstitutes to a dosage strength of 10 mcg/ml. Label the contents.
- Withdraw 5 ml of the drug and administer 1 ml every 3-5 minutes checking B/P prior to administration until a B/P of 90 systolic is obtained. Label the syringe. Once B/P is achieved, place IVF at KVO. Monitor closely and repeat procedures in accordance with protocol.

Epinephrine 1:10,000: *Protocol: C-7 PEA/Asystole*

Protocol: C-9 Ventricular Fibrillation/Pulseless Ventricular Tachycardia

Administer **epinephrine 1:10,000 concentration** 1 mg IV or IO. Repeat every 3- 5 minutes prn cardiac arrest.

Fentanyl: *Protocol: G-5 Pain Management Non-Cardiac/Procedural Analgesis*

Fentanyl 50 mcg IVP slowly or IN (25 mcg per nostril) for severe pain if:

GCS ≥ 14

Pain scale > 5

B/P ≥ 90 systolic

May be repeated every 10 minutes as long as the above parameters are still met.

Additional doses required before the 10 minute interval are AT MD DISCRETION ONLY.

Protocol: C-4 Chest Pain Believed to be Cardiac in Origin

Fentanyl may be administered for documented, continuing pain > 5 on the pain scale AFTER the paramedic administration of 3 nitroglycerin

Fentanyl 25 mcg IVP slowly over 1-2 minutes. May repeat every 10 minutes at same dose as long as pain > 5 on pain scale and B/P > 90 systolic.

Protocol G-10 Post Intubation Sedation and Pain Control

For agitation, Administer 100 mcg Fentanyl IV/IO. If B/P > 90. This dosing regimen may be repeated every 5 minutes as needed.

Ketamine: *Protocol G-8 Ketamine for Acute Pain Control*

Ketamine 0.2 mg/Kg IV push over 90 seconds, repeat in 10 minutes if conditions are met.

Protocol G-2 Ketamine Behavioral Emergencies/Restraint

For chemical restraint: For Adult only. Adult <80 kg: administer 300 mg in the vastus lateralis. Adult > 80 kg: administer 400 mg in the vastus lateralis.

Protocol G-10 Post Intubation Sedation and Pain Control

Ketamine 1 mg/kg IV/IO SLOWLY May be given INITIALY: This dose may be repeated in 15min if needed one time.

Lidocaine 2% gel: *Protocol: A-1 Airway Management*

Lubricate a nasopharyngeal airway with 2% Xylocaine gel and insert it into the larger nostril if time allows prior to the intubation. You may further lubricate the ET tube with the 2% lidocaine gel prior to insertion.

Lidocaine 2%: *Protocol: C-11 Sustained Undifferentiated Wide Complex Tachycardia*

If lidocaine is ordered, "AT MD DISCRETION ONLY" administer **Lidocaine 1mg/Kg IVP** slowly at a rate of 50 mg/min.

Protocol: O-2 Suspected Overdose/Poisoning

For VT in tricyclic antidepressant overdose, administer **Lidocaine 1.5 mg/Kg IVP.**

Protocol: G-1 Alternate Venous Access Devices/intraosseous Insertion

In the adult patient administer **20-50 mg of Lidocaine 2% SLOWLY IO** prior to fluid administration. May repeat once for continued discomfort.

Narcan: *Protocol: O-2 Suspected Overdose/Poisoning*

For suspected opiate overdose, administer naloxone 2 mg IN, 1 mg in each nostril OR administer naloxone 0.5 mg IV.

Neo-synephrine: *Protocol: A-2 Airway Management*

For nasotracheal intubation, apply two or three sprays of NeoSynephrine spray in each nostril to vasoconstrict nasal tissue.

Nitroglycerin: *Protocol: C-4 Chest Pain Believed to be Cardiac in Origin*

May administer nitroglycerin 0.4 mg SL repeating every 3-5 minutes up to three doses total as long as pain persists and B/P is maintained above 110.

Protocol: C-5 Congestive Heart Failure/Pulmonary Edema

If systolic B/P systolic is ≧ 110 but < 160 and diastolic is normal or elevated:

Administer 1 dose of NTG 0.4 mg SL and repeat every 5 minutes up to six doses.

If B/P > 160/ and diastolic is normal or elevated

Administer 3(three) NTG 0.4 mg SL as one dose. Repeat dose in 5 minutes unless B/P drops below 160/90

Solu Medrol: *Protocol: M-7 Respiratory Distress with/ without wheezing*

125mg IVP slowly over one (1) minute.

Protocol: M-2 Allergic Reaction/Anaphylaxis

If epinephrine is administered, Solu Medrol 125mg IVP slowly.

Sodium Bicarbonate 8.4% *Protocol: O-2 Suspected Overdose/Poisoning*

In Tricyclic antidepressant overdose, If patient develops a wide complex rhythm (QRS > 0.12 seconds) or VT OR there is hypotension refractory to 500 ml of fluids, administer sodium bicarbonate 1 meq/Kg IVP.

Protocol: C-7 PEA/Asystole

HYPERKALEMIA: Administer 8.4% Sodium Bicarbonate 50 mEq IVP/IO

Tetracaine: Utilized by Tactical Rescue paramedic only

Tranexamic Acid: Protocol G-11

Administer 1 G TXA IV/IO by:

Mixing 1 G TXA into 100 ml NS and run it in over 10 minutes.

Use dedicated line if possible. Do not administer in same line as blood or blood products, Factor VIIa, or Penicillin.

Versed: *Protocol: G-5 Pain Management Non-cardiac/Procedural Analgesia*

For synchronized cardioversion/pacing, Versed: 10 mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if GCS ≥ 14

Protocol: M-8 Seizure

Versed 10 mg (2ml) IN placing 5 mg (1ml) in each nostril, or

Versed 2.5 mg-5 mg IVP slowly, or

Versed 10 mg IM in the vastus lateralis

May repeat either dose every 5 minutes for continued seizure activity. *Maximum cumulative dose 20 mg.*

Doses above 20 mg AT MD DISCRETION ONLY

Protocol: O-2 Suspected Overdose/Poisoning

For extremely agitated patient under the influence of cocaine, ecstacy, amphetamine OD administer Versed: 10 mg IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP. May repeat either dosing to a maximum cumulative dose of 25 mg. Call Medical Control for further dosing as needed.

If the rhythm is SVT or VT that is hemodynamically stable in a cocaine, ecstacy, amphetamine OD with or without symptoms, administer Versed: 10 mg (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP.

Protocol: G-2 Behavioral Emergencies/Restraint

For chemical restraint, Adult only > 50 KG: Administer 10 mg (2 ml) of midazolam utilizing a concentration of 5 mg/ml IN (intranasal) with the nasal atomizer. Place 1 ml in each nostril.

. Dosing may be repeated every 5 minutes until patient's BARS score improves.

Protocol: G-8 Ketamine for Acute Pain Control

Laryngospasm, administer 2mg of midazolam IVP every 3 minutes.

Emergence Reaction: administer 2-5 mg midazolam over 2 minutes

Protocol G-10 Post Intubation Sedation and Pain Control

Versed 5 mg IV/IO, may repeat once in 10 minutes

Zofran: *Protocol: M-1 Abdominal Pain*

Nausea and/or vomiting, administer Zofran (ondansetron hydrochloride) 4 mg undiluted IVP SLOWLY over 2-5 minutes. May repeat in 10 minutes if needed. If IV is not available, may administer Zofran 4 mg IM in the vastus lateralis muscle or deltoid.

Protocol: O-2 Suspected Overdose/Poisoning

Fore green tobacco sickness or nicotine poisoning, Administer Zofran 4 mg IV for nausea and vomiting. May repeat once in 10 minutes at 4 m

Protocol: G-5 Pain Management/Procedural Analgesia

Administer Zofran 4 mg slow IVP as needed for nausea and vomiting.

Protocol: G-8 Ketamine for Acute Pain Control

Administer Zofran per Abdominal Pain protocol.

Protocol: M-4 Heat Emergencies

Nausea and/or vomiting, administer Zofran (ondansetron hydrochloride) 4 mg undiluted IVP SLOWLY over 2-5 minutes. May repeat in 10 minutes if needed. If IV is not available, may administer Zofran 4 mg IM in the vastus lateralis muscle or deltoid.